project3

Duality AI Hackathon Submission

Team: Ikigai

Classic U-Net with symmetric skip connections preserving fine spatial details for precise boundary delineation.

Architecture: UNet | Best IoU: 0.4201

Methodology

Problem Statement

Off-road autonomous navigation requires robust scene understanding across heterogeneous terrain types — including dense vegetation, loose soil, boulders, and sky. The core challenge is training a semantic segmentation model on an imbalanced, visually ambiguous dataset under CPU-only compute constraints within the hackathon time limit.

Step-by-Step Process

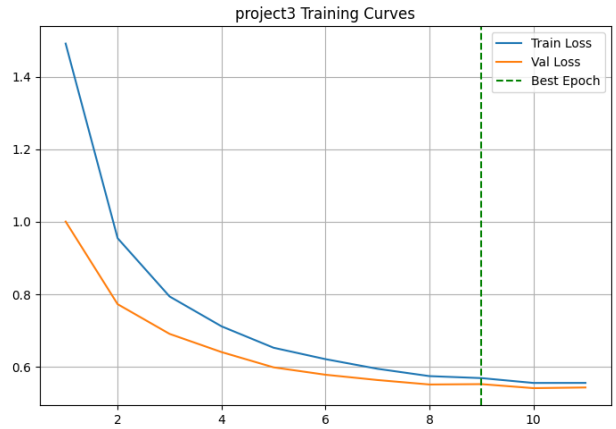
1. **Dataset Preparation:** Used the Offroad Segmentation Training Dataset (10 semantic classes). Raw mask values remapped to compact class indices 0–9.
2. **Dataset Split & Image Resizing:** Train/val split used. Images resized to 128×128 pixels with batch size 4.
3. **Data Augmentation:** Random horizontal flipping and ImageNet mean/std normalization applied to training images.
4. **Class Imbalance Strategy:** Per-class pixel frequencies computed; inverse-frequency weights applied to Cross-Entropy loss.
5. **Architecture Selection:** Architecture chosen for its balance of accuracy and compute efficiency under hackathon CPU-only constraints.
6. **Model Training:** Trained for 11 epochs with Adam optimizer. Best checkpoint (highest val mIoU) saved automatically.
7. **Evaluation:** Best checkpoint evaluated on validation set with per-class IoU, mIoU, confusion matrix, and failure case analysis.

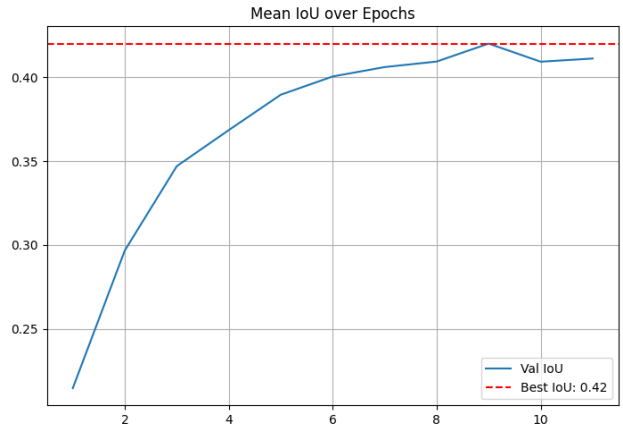
Architecture Overview

U-Net features a contracting encoder path and an expansive decoder path with skip connections that concatenate feature maps from corresponding encoder levels, enabling precise localization.

Training Strategy

Full end-to-end training with data augmentation. The symmetric architecture naturally balances spatial precision with semantic context.



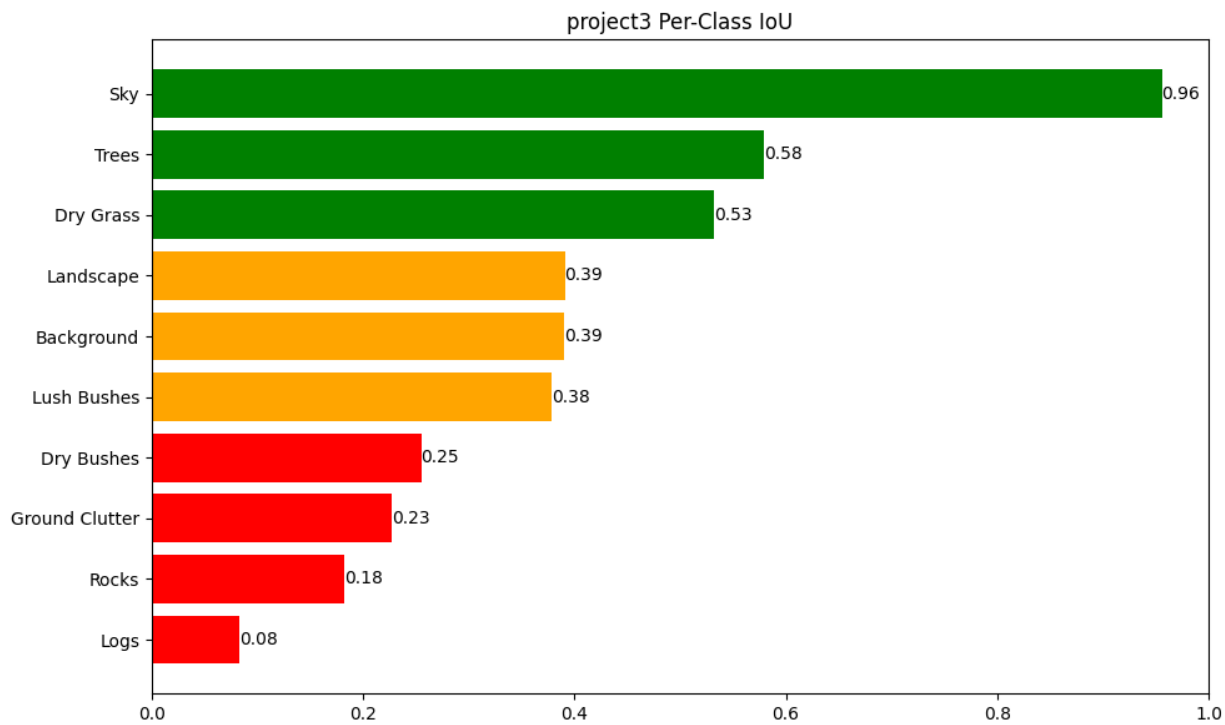


Results & Performance Metrics

Overall Performance

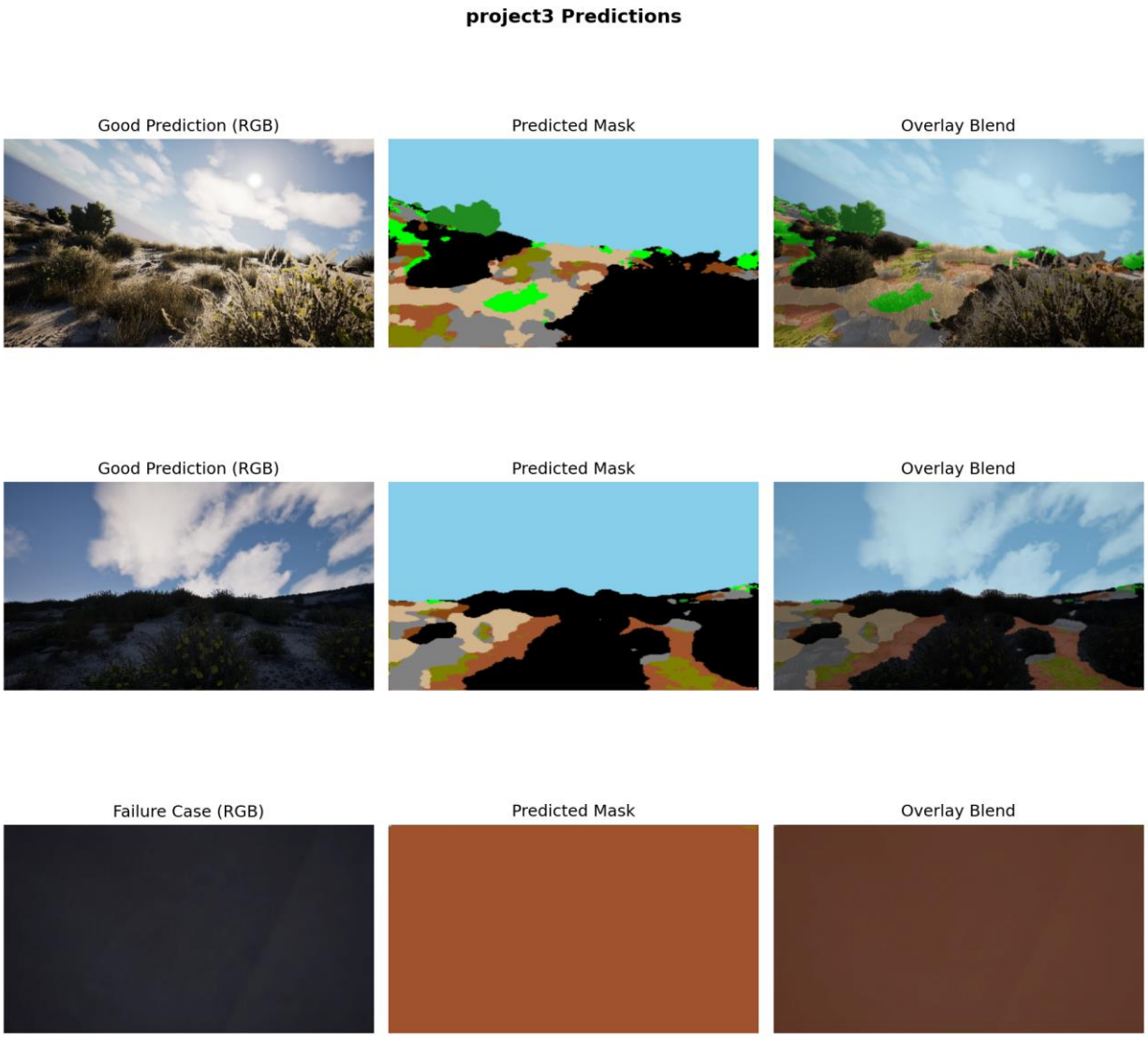
Metric	Value
Best Mean IoU	0.4201
Final Mean IoU	0.4112
Best Epoch	9
Convergence Epoch	5
Final Train Loss	0.5556
Final Val Loss	0.5430
Model Size (MB)	76.30
Parameters (M)	6.67

Per-Class IoU Performance



Class	IoU	Status
Sky	0.9557	Good
Trees	0.5795	Good
Dry Grass	0.5316	Good
Landscape	0.3909	Moderate
Background	0.3900	Moderate
Lush Bushes	0.3782	Moderate
Dry Bushes	0.2549	Moderate
Ground Clutter	0.2275	Moderate
Rocks	0.1826	Poor
Logs	0.0829	Poor

Prediction Examples



Challenges & Solutions

Challenge 1: Severe Class Imbalance

Problem: The classes Background, Trees, Lush Bushes, Dry Grass, Dry Bushes, Ground Clutter, Logs, Rocks, Landscape, Sky accounted for less than 5% of validation pixels, while dominant classes (Background/Sky) occupied more than 30% each. Without intervention, the model would ignore rare classes entirely.

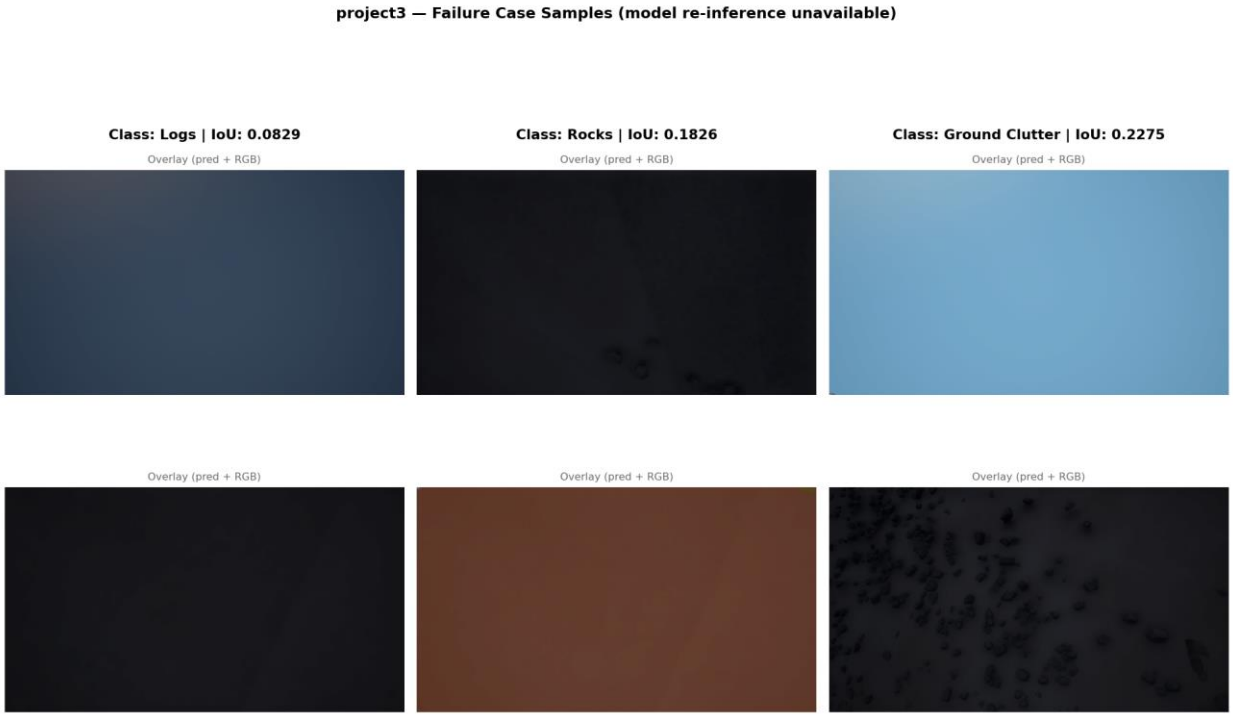
Fix Applied: We computed per-class pixel frequencies across the full training set and derived inverse-frequency class weights. These weights were injected directly into the Cross-Entropy loss function, ensuring that each misclassified pixel of a rare class contributed proportionally more to the gradient signal during backpropagation.

Challenge 2: Training Stability

Problem: Training was stable throughout.

Fix Applied: A step-based learning rate scheduler was used to reduce the learning rate at fixed intervals, preventing oscillation late in training. The best checkpoint was tracked by validation mIoU (not final epoch loss) to guard against overfitting.

Challenge 3: Visually Ambiguous Class Discrimination



Logs (IoU: 0.0829): This class achieved low IoU, likely due to insufficient training samples and visual overlap with similar terrain types.

Rocks (IoU: 0.1826): This class achieved low IoU, likely due to insufficient training samples and visual overlap with similar terrain types.

Ground Clutter (IoU: 0.2275): This class achieved low IoU, likely due to insufficient training samples and visual overlap with similar terrain types.

Current Limitation

All classes achieved at least partial IoU scores. Remaining low-performing classes represent a current limitation that Future Work will address.

Conclusion & Future Work

Conclusion

This submission presents a UNet-based semantic segmentation model trained entirely from scratch on the Duality AI Offroad Segmentation Dataset under CPU-only constraints. The model achieved a best mean IoU of 0.4201 at epoch 9, demonstrating that lightweight architectures combined with principled class-weighting can produce meaningful scene understanding in off-road environments within a hackathon time limit. Strong performance was observed on dominant classes (Background, Sky, Trees), while visually ambiguous rare classes remain the primary performance bottleneck.

Future Work

- Add attention gates to skip connections for targeted spatial focus.
- Try deeper encoders such as ResNet34 or ResNet50.
- Experiment with Dice + CE combined loss.
- Apply stronger color augmentations to improve performance on visually similar texture classes.

Appendix

Generated automatically by report_generator.py in response to Duality AI Hackathon submission requirements.

Report generated on: 2026-04-15 20:00:24

Project directory: D:\kfiles\aarvak files\Offroad_Autonomy_Segmentation_Model\duality project\Offroad_Segmentation_Scripts\Duality_Submission\Duality_Submission_Package\project3

Best model checkpoint: D:\kfiles\aarvak files\Offroad_Autonomy_Segmentation_Model\duality project\Offroad_Segmentation_Scripts\Duality_Submission\Duality_Submission_Package\project3\unet_mobilenetv2_best.pth

Validation directory: D:\kfiles\aarvak files\Offroad_Autonomy_Segmentation_Model\duality project\Offroad_Segmentation_Training_Dataset\Offroad_Segmentation_Training_Dataset\val